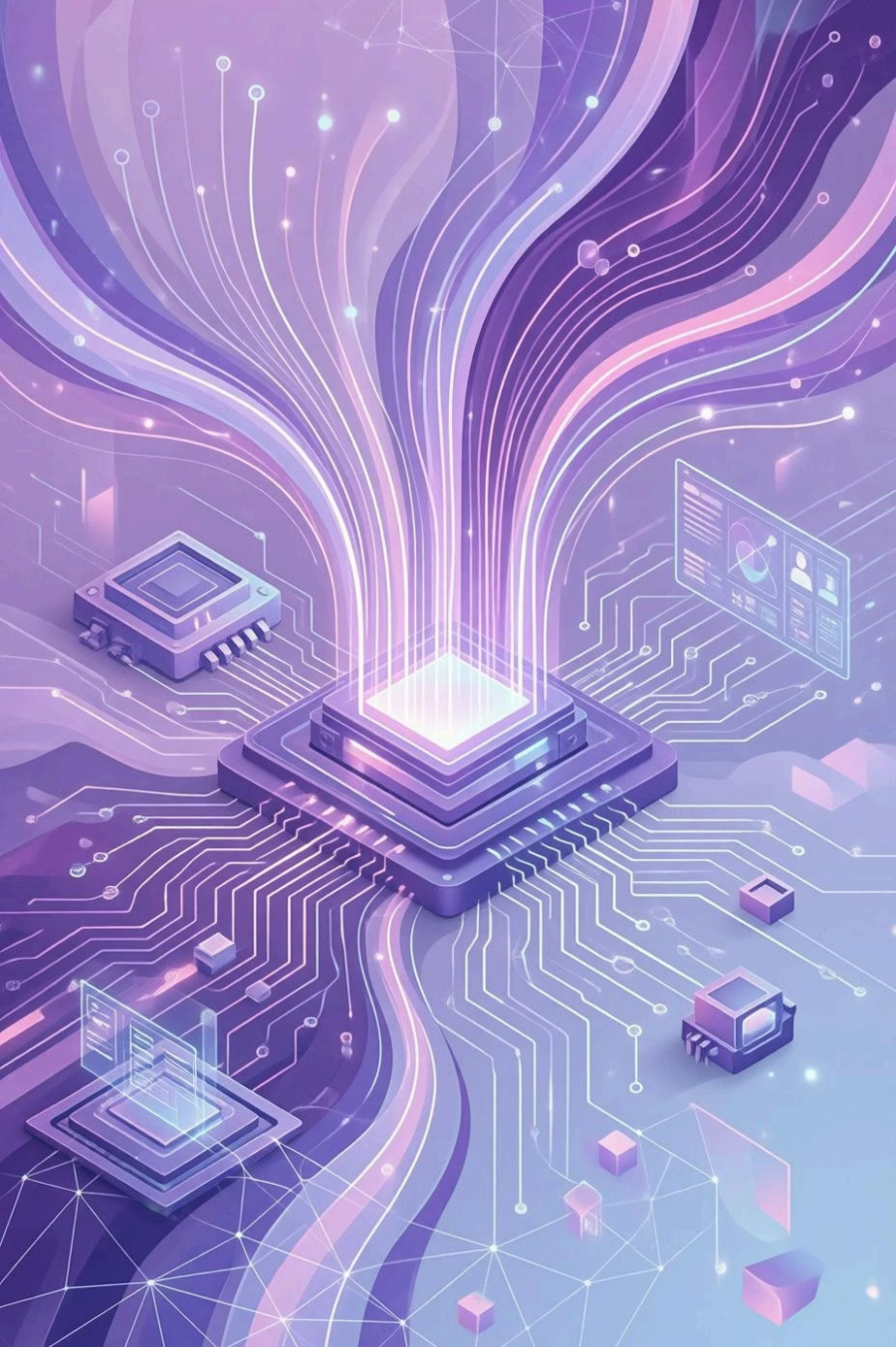




IoT FPS Controller & Real-Time Web Game

A Multi-Tier Architecture Demonstration

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ESP32

</> PYTHON

LARAVEL 12

THREE.JS

Project Goal & Vision

The Primary Objective

Design and build a full-stack, multi-tier IoT application that bridges the physical world with a digital web environment in real-time.

The Concept

A custom-built "Gun Controller" enabling players to aim and move inside a web-based 3D First-Person Shooter game through natural physical gestures.

Low Latency

Achieving sub-50ms response time for smooth, competitive gaming experience without perceptible input lag.

Sensor Fusion

Combining Gyroscope and Joystick inputs to create intuitive, natural control that feels like an extension of the player's body.

Bi-Directional Communication

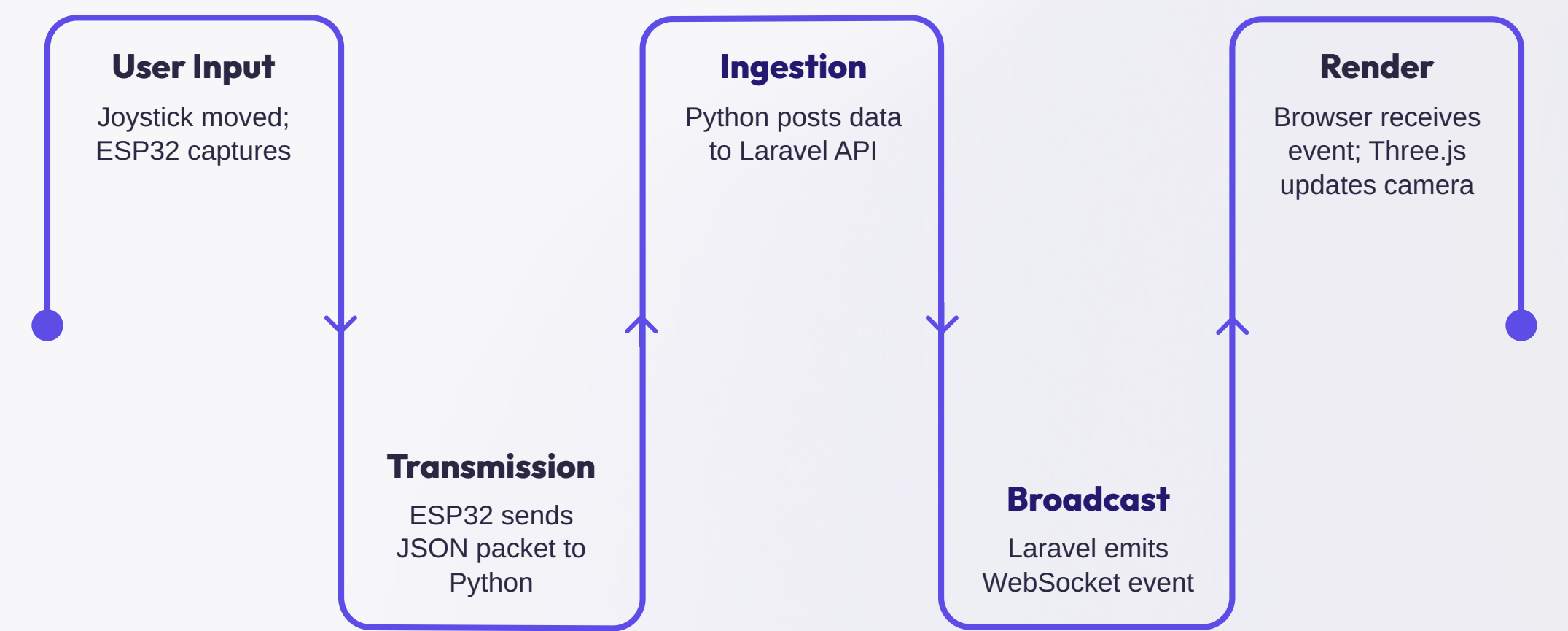
Sending precise controls to the game engine while receiving instant audio feedback to the physical controller.

System Architecture

The 4-Tier Model



Data Flow Journey



From physical gesture to visual response, data travels through five optimized stages, each designed to minimize latency and maximize responsiveness.

Hardware & Software Implementation



The Controller

ESP32 WROOM (Dual-core 240MHz) serves as the brain, coordinating multiple sensors and providing real-time feedback.

- GY-521 (MPU-6050):** 6-Axis Gyroscope/Accelerometer for precise aiming control (Pitch/Yaw)
- Analog Joystick:** Character locomotion with full 360° movement
- Passive Buzzer:** Localized audio feedback synchronized with on-screen actions

1

Hardware Oversampling

Reads sensors 5 times per frame and averages results to eliminate electrical noise and jitter.

2

Input Latching

Holds trigger signal for 60ms ensuring server never misses a shot due to network timing variance.

3

Audio Sync Delay

150ms buzzer delay perfectly matches visual muzzle flash on screen for immersive feedback.

Backend Architecture

Laravel 12 powers the authoritative game server with optimized API routes accepting compressed JSON payloads (GX, GY, JX, JY, F, C).

The **ControllerDataEvent** broadcasts data immediately using [Laravel Reverb](#), bypassing queue delays for true real-time communication.

Frontend Engine

Three.js handles sophisticated input processing:

- Rate Integration:** Converts Gyro rates (°/sec) into smooth camera rotation
- Weighted Moving Average:** 2-frame history buffer filters hand tremor while maintaining responsiveness
- Circular Deadzone:** Magnitude-based calculation enables precise omnidirectional movement